Notes: Jarosch, Oberfield & Rossi-Hansberg (Econometrica, 2021)

Learning From Coworkers

Contents

1 Big picture

- **Question.** Do workers learn from coworkers, and how large is the market value of this learning?
- **Setting.** The paper uses German linked employer-employee data with the full workforce of sampled establishments. This lets the authors observe a worker's wage, team, occupation, and coworker wage distribution.
- **Reduced-form fact.** A worker's future wage growth is strongly related to the wages of current coworkers. The relationship is much stronger for coworkers paid more than the worker than for coworkers paid less.
- **Narrow-team result.** Effects are larger when teams are defined as establishment-by-occupation groups than when teams are defined as whole establishments. This suggests learning is more intense among coworkers who use related knowledge.
- **Robustness.** The pattern survives checks on worker switchers, mass-layoff movers, tenure groups, age groups, wage deciles, occupation splits, establishment wage-bill growth, and employment growth.
- **Theory.** The authors build a competitive labor-market model where workers produce in teams and learn from coworkers. Wages are not just pay for current productivity; they also reflect compensating differentials for learning opportunities.
- **Structural idea.** In a competitive market, a worker's value equals wages today plus the expected value of knowledge tomorrow. Choosing knowledge units as expected present value of earnings lets the authors infer latent knowledge from wages and team composition.
- **Main structural result.** Learning from higher-knowledge coworkers is much stronger than learning from lower-knowledge coworkers. The estimated effect is larger for narrow occupation teams.
- **Compensation result.** Coworker learning is economically large: roughly 4–9 percent of total worker compensation is received as knowledge accumulation rather than current wages.
- **Sorting result.** Randomly reshuffling workers across teams would raise knowledge growth by roughly 64–91 percent, suggesting equilibrium production sorting limits learning opportunities.
- **Inequality result.** Once learning compensation is included, inequality is smaller than wage inequality. The variance of log compensation is about one-fifth to one-third lower than the variance of log wages.
- **Takeaway.** Coworkers are a major source of human-capital accumulation. Team formation affects both production today and the diffusion of knowledge tomorrow.

2 Data and measurement

2.1 German matched employer-employee data

- The data are German administrative social-security records linked to establishments.
- The sample covers 1999–2010 and includes the full workforce of sampled establishments.
- Observed variables include establishment, occupation, daily earnings, age, gender, tenure, education, location, and other worker characteristics.
- The paper uses prime-age, full-time workers and trims extreme wage-growth observations in reduced-form regressions.

Interpretation: The data are unusually useful because the authors can observe not only a worker's own wage history, but also the full wage distribution of the coworkers with whom the worker interacts.

2.2 Team definitions

The paper uses two team definitions.

- **Team Definition 1:** all workers in the same establishment.
- **Team Definition 2:** all workers in the same establishment and occupation.

Interpretation: Team Definition 2 is narrower and closer to daily production interactions. Larger effects under this definition are interpreted as evidence that relevant learning is more local within occupations.

2.3 Wage variables

For worker i in year t:

- $w_{i,t}$ is log wage;
- $\bar{w}_{-i,t}$ is the log mean wage of coworkers;
- $\bar{w}^{+}_{-i,t}$ is the log mean wage of coworkers paid more than worker i;
- $\bar{w}^{-}_{-i,t}$ is the log mean wage of coworkers paid less than worker i.

Interpretation: Coworker wages proxy for coworker knowledge. The key learning prediction is that higher-knowledge coworkers should increase future wage growth, especially if they are close enough to teach relevant knowledge.

2.4 Controls

The reduced-form regressions include:

- current own wage;
- age-decile fixed effects;
- tenure-decile fixed effects;
- gender fixed effects;
- education fixed effects;
- occupation fixed effects;
- year fixed effects.

Interpretation: Controlling for current own wage is critical because wage growth is compared among workers with similar current wage levels but different coworker wage distributions.

2.5 Structural objects

The structural model uses:

- z_i: worker i's knowledge;
- $\widetilde{z}_{-i}$: vector of coworker knowledge;
- $G(z' \mid z, \widetilde{z})$: law of motion for knowledge after working with coworkers;
- $V(z)$: expected present value of earnings;
- β: discount factor, set externally in estimation.

Interpretation: The paper chooses the cardinality of z so that $V(z) = z$. Knowledge is measured in units of expected present value of earnings.

3 Models and empirical specifications

3.1 Reduced-form baseline

The first regression is estimated separately for horizons h:

$$w_{i,t+h} = \alpha + \beta \bar{w}_{-i,t} + \gamma w_{i,t} + \omega_{age} + \omega_{tenure} + \omega_{gender} + \omega_{educ} + \omega_{occ} + \omega_t + \varepsilon_{i,t}. \tag{1}$$

Interpretation: A positive β means workers with higher-paid coworkers today have higher wages in the future, conditional on their own current wage and controls.

3.2 High-paid and low-paid coworker split

The main reduced-form specification separates coworkers above and below worker i:

$$w_{i,t+h} = \alpha + \beta^+ \bar{w}^+_{-i,t} + \beta^- \bar{w}^-_{-i,t} + \gamma w_{i,t} + \omega_{age} + \omega_{tenure} + \omega_{gender} + \omega_{educ} + \omega_{occ} + \omega_t + \varepsilon_{i,t}. \tag{2}$$

Interpretation: The learning interpretation predicts $\beta^+ > \beta^-$. Higher-paid coworkers should contain more useful knowledge for the worker.

3.3 Flexible coworker wage distribution

The paper bins coworkers by the wage gap $\log(w_j) - \log(w_i)$ and estimates

$$w_{i,t+h} = \sum_{k=2}^{11} \beta_k p_{i,k,t} + \gamma w_{i,t} + \omega_{age} + \omega_{tenure} + \omega_{gender} + \omega_{educ} + \omega_{occ} + \omega_t + \varepsilon_{i,t}, \tag{3}$$

where $p_{i,k,t}$ is the fraction of coworkers in bin k.

Interpretation: This specification asks whether the whole shape of the peer wage distribution matters, not just the mean wage of workers above and below.

3.4 Worker value and wage equation

In the model, a worker with knowledge z chooses coworkers $\widetilde{z}$ to maximize

$$V(z) = \max_{\widetilde{z} \in \widetilde{Z}} \left\{ w(z; \widetilde{z}) + \beta \int V(z') dG(z' \mid z, \widetilde{z}) \right\}. \tag{4}$$

In competitive equilibrium, wages satisfy

$$w(z; \widetilde{z}) = V(z) - \beta \int V(z') dG(z' \mid z, \widetilde{z}). \tag{5}$$

Interpretation: A firm that provides better learning opportunities can pay a lower current wage and still attract the worker. Wages contain compensating differentials for learning.

3.5 Knowledge units and Bellman inversion

The authors choose units of knowledge so that $V(z) = z$. The worker equation becomes

$$z_i = w_i + \beta E[z'_i \mid z_i, \widetilde{z}_{-i}]. \tag{6}$$

Interpretation: Given a learning function, this equation can be inverted within each observed team to infer every worker's latent knowledge from wages and coworker composition.

3.6 GMM moment conditions

Write next-period knowledge as

$$z_i' = E(z_i, \widetilde{z}_{-i}) + \varepsilon_i. \tag{7}$$

If the conditional expectation is linear in moments $m_k(z_i, \widetilde{z}_{-i})$, the orthogonality conditions are

$$E\left[m_k(z_i, \widetilde{z}_{-i})\varepsilon_i\right] = 0, \qquad k = 1, \ldots, K. \tag{8}$$

Interpretation: The estimation alternates between recovering latent knowledge from wages and updating learning-function parameters using the implied panel of knowledge.

3.7 Baseline structural learning function

The baseline learning function is

$$E[z_i' - z_i \mid z_i, \widetilde{z}_{-i}] = \theta^0 z_i + \frac{1}{n-1}\left\{\theta^- \sum_{z_j < z_i}(z_j - z_i) + \theta^+ \sum_{z_j \geq z_i}(z_j - z_i)\right\}. \tag{9}$$

Interpretation: θ^+ measures learning from more-knowledgeable coworkers; θ^- measures the effect of less-knowledgeable coworkers; θ^0 is trend knowledge growth.

3.8 Piecewise linear learning function

The richer function allows the marginal effect of higher-knowledge coworkers to change once the gap is large:

$$\Theta'(x) = \begin{cases} \theta^{++}, & x \geq 1 + b, \\ \theta^+, & 1 \leq x < 1 + b, \\ \theta^-, & x < 1. \end{cases} \tag{10}$$

Interpretation: The estimate $\theta^+ > \theta^{++}$ means that learning from better coworkers is strongest when the better coworker is not too far above the worker in the knowledge distribution.

3.9 Team-size learning function

To study scale effects, the paper estimates

$$E[z_i' - z_i \mid z_i, \widetilde{z}_{-i}] = \theta^0 z_i + \frac{1}{(n-1)^k}\left\{\theta^- \sum_{z_j < z_i}(z_j - z_i) + \theta^+ \sum_{z_j \geq z_i}(z_j - z_i)\right\}. \tag{11}$$

Interpretation: If $k = 1$, learning is averaged over coworkers and has constant returns to scale. If $k < 1$, larger teams have increasing returns to learning.

4 Identification and empirical design

4.1 Reduced-form identification logic

- Workers with similar current wages are compared across different coworker wage distributions.
- The key coefficient asks whether the wage gap to coworkers predicts future wage growth.

- Asymmetry between high-paid and low-paid coworkers is central: learning predicts stronger effects from higher-paid coworkers.
- The switcher and mass-layoff analyses check whether the relationship is purely establishment-specific or back-loaded pay.

Interpretation: The reduced-form results are not a randomized experiment, but the pattern across many cuts makes learning a parsimonious explanation.

4.2 Alternative explanations addressed

- **Back-loaded wages:** results persist for job switchers and workers leaving after mass layoffs.
- **Mean reversion:** the high/low peer asymmetry and Figure 1 are hard to reconcile with simple mean reversion.
- **Rent-sharing:** establishment wage-bill and employment-growth controls do not change the key results.
- **Worker sorting:** results are similar when controlling for previous wage growth.
- **Firm-level learning shocks:** team fixed-effect and establishment growth checks limit this concern.

Interpretation: No single robustness check proves causality, but the whole set narrows the set of plausible alternatives.

4.3 Structural identification logic

- The structural approach relies on the worker Bellman equation, not on restrictions on firms' production functions.
- Competitive labor markets imply that wage differences compensate workers for current productivity and learning opportunities.
- The authors infer knowledge from wages and team composition for year t and $t + 1$.
- Parameters are chosen so that inferred knowledge evolves consistently with the learning function.

Interpretation: The key methodological contribution is that learning can be estimated using observed wages and teams without specifying the full firm production technology.

4.4 Limitations

- The reduced-form estimates are equilibrium associations except for the noisier pseudo-random exit evidence in the appendix.
- The structural model assumes stationarity, competitive labor markets, and complete markets or linear utility.
- Wages are assumed to reflect knowledge and learning differentials; rent-sharing and search frictions are discussed as extensions.
- The model recovers knowledge only up to the chosen cardinality of expected present value of earnings.

Interpretation: The model is best read as a benchmark that turns the wage-team relationship into an interpretable estimate of knowledge flows.

5 Tables and figures

This section follows the compact format: adjacent tables and figures are grouped to save space. All visuals are directly cropped from the original paper.

5.1 Tables I–III: baseline reduced-form wage-growth evidence

Read:

- Table I estimates the baseline coworker mean-wage specification. Doubling peers' wages raises next-year wages by about 6–7 percent and ten-year wages by about 21 percent.
- The effect is somewhat larger for the narrow establishment-by-occupation team definition at short horizons.
- Table II splits coworkers above and below the worker. At the one-year horizon, $\bar{w}^+$ has coefficient 0.090 while $\bar{w}^-$ has coefficient 0.019.
- At the ten-year horizon, the high-paid peer coefficient is 0.28 while the low-paid peer coefficient is 0.097.
- Table III shows the same asymmetric pattern using the broad establishment team definition.

Economic message: Higher-paid coworkers matter much more for future wages than lower-paid coworkers, which is the core reduced-form evidence for learning.

TABLE I

ESTIMATION RESULTS FOR SPECIFICATION (1)[a]

	Narrow Team Definition				
Horizon in Years	1	2	3	5	10
$\bar{w}$	0.068 (0.0036)	0.094 (0.0052)	0.12 (0.0071)	0.16 (0.0094)	0.21 (0.014)
Within R^2	0.89	0.82	0.77	0.68	0.47
Observations	3,969,166	3,473,642	2,989,524	2,167,851	508,504

	Broad Team Definition				
Horizon in Years	1	2	3	5	10
$\bar{w}$	0.058 (0.0031)	0.083 (0.0046)	0.11 (0.0063)	0.15 (0.0086)	0.21 (0.014)
Within R^2	0.89	0.83	0.78	0.68	0.48
Observations	4,112,284	3,595,957	3,092,504	2,239,223	523,683

[a]*Notes*: $\hat{\beta}$ as estimated from specification (1). Column titles indicate horizon n. Standard errors clustered at the establishment level. The regressions include current wage and fixed effects for age decile, tenure decile, gender, education, occupation, and year. Standard errors in parentheses.

Table I: mean coworker wage.

TABLE II

ESTIMATION RESULTS FOR SPECIFICATION (2)[a]

	Horizon in Years				
	1	2	3	5	10
$\bar{w}^+$	0.090 (0.0064)	0.13 (0.010)	0.16 (0.015)	0.22 (0.021)	0.28 (0.032)
$\bar{w}^-$	0.019 (0.0048)	0.029 (0.0065)	0.041 (0.0081)	0.060 (0.011)	0.097 (0.016)
Within R^2	0.88	0.81	0.76	0.66	0.46
Observations	3,462,305	3,034,301	2,617,097	1,903,104	448,560

[a]*Notes*: $\hat{\beta}^+$ and β^- as estimated from specification (2). Team Definition 2. Column titles indicate horizon n. Standard errors clustered at the establishment level. The regressions include current wage and fixed effects for age decile, tenure decile, gender, education, occupation, and year. Standard errors in parentheses.

TABLE III

ESTIMATION RESULTS FOR SPECIFICATION (2)—TEAM DEFINITION 1[a]

	Horizon in Years				
	1	2	3	5	10
$\bar{w}^+$	0.090 (0.0058)	0.13 (0.0089)	0.17 (0.012)	0.23 (0.017)	0.32 (0.026)
$\bar{w}^-$	0.025 (0.0044)	0.039 (0.0060)	0.057 (0.0081)	0.085 (0.012)	0.12 (0.019)
Within R^2	0.89	0.82	0.77	0.68	0.48
Observations	4,026,321	3,522,994	3,032,228	2,197,932	515,017

[a]*Notes*: $\hat{\beta}^+$ and β^- as estimated from specification (2). Team Definition 1. Column titles indicate horizon h. Standard errors clustered at the establishment level. The regressions include current wage and fixed effects for age decile, tenure decile, gender, education, occupation, and year. Standard errors in parentheses.

Tables II–III: high-paid and low-paid coworker split.

5.2 Table IV: heterogeneity by wage, age, tenure, and team size

Read:

- Panel A shows the high-paid coworker effect across own-wage deciles. The effect remains positive throughout and is especially large at the top of the wage distribution.
- Panel B shows learning from high-paid coworkers is strongest for younger workers.
- Panel C shows the high-paid coworker effect is strongest at low tenure and declines with tenure.

- Panel D shows the pattern is present across team-size deciles, with less evidence that the effect disappears in large teams.

Economic message: The effects align with learning: workers with more room to learn - younger and lower-tenure workers - respond more to high-paid coworkers.

TABLE IV

BASELINE RESULTS FOR DIFFERENT DECILES OF THE WAGE, AGE, TENURE, AND TEAM SIZE DISTRIBUTION[a]

	Panel A: Decile of the Wage Distribution									
	1	2	3	4	5	6	7	8	9	10
$\bar{w}^+$	0.18 (0.012)	0.13 (0.015)	0.14 (0.018)	0.13 (0.018)	0.13 (0.021)	0.11 (0.026)	0.11 (0.028)	0.10 (0.023)	0.32 (0.021)	0.35 (0.060)
$\bar{w}^-$	0.043 (0.0097)	0.041 (0.010)	0.045 (0.012)	0.047 (0.014)	0.048 (0.015)	0.046 (0.015)	0.071 (0.015)	0.069 (0.0096)	0.057 (0.010)	0.024 (0.0058)
Within R^2	0.43	0.086	0.053	0.040	0.037	0.039	0.051	0.080	0.17	0.062
Observations	248,920	263,528	265,462	264,823	262,632	261,579	261,499	261,687	262,796	264,087

	Panel B: Decile of the Age Distribution									
	1	2	3	4	5	6	7	8	9	10
$\bar{w}^+$	0.31 (0.021)	0.27 (0.026)	0.16 (0.017)	0.12 (0.014)	0.097 (0.013)	0.089 (0.012)	0.058 (0.010)	0.035 (0.0092)	0.018 (0.0093)	0.0092 (0.0088)
$\bar{w}^-$	0.023 (0.014)	0.032 (0.011)	0.030 (0.0089)	0.038 (0.0084)	0.045 (0.0088)	0.046 (0.0083)	0.056 (0.0082)	0.061 (0.0073)	0.065 (0.0079)	0.056 (0.0069)
Within R^2	0.61	0.70	0.75	0.77	0.78	0.79	0.80	0.82	0.82	0.82
Observations	313,804	256,870	253,754	294,113	205,242	301,874	284,811	261,404	221,992	223,184

	Panel C: Decile of the Tenure Distribution									
	1	2	3	4	5	6	7	8	9	10
$\bar{w}^+$	0.26 (0.023)	0.25 (0.027)	0.19 (0.025)	0.16 (0.017)	0.10 (0.014)	0.077 (0.014)	0.062 (0.012)	0.050 (0.013)	0.035 (0.014)	0.012 (0.014)
$\bar{w}^-$	0.012 (0.011)	0.022 (0.0086)	0.038 (0.0090)	0.059 (0.0091)	0.051 (0.0085)	0.061 (0.0093)	0.062 (0.0089)	0.065 (0.010)	0.050 (0.013)	0.059 (0.016)
Within R^2	0.66	0.73	0.77	0.79	0.80	0.79	0.79	0.78	0.74	0.74
Observations	259,787	258,018	261,847	264,724	314,261	216,646	298,387	231,182	261,285	250,908

	Panel D: Decile of the Size Distribution									
	1	2	3	4	5	6	7	8	9	10
$\bar{w}^+$	0.057 (0.0049)	0.097 (0.0081)	0.13 (0.013)	0.16 (0.017)	0.16 (0.020)	0.23 (0.027)	0.24 (0.043)	0.25 (0.054)	0.16 (0.040)	0.16 (0.11)
$\bar{w}^-$	0.027 (0.0036)	0.040 (0.0063)	0.050 (0.0080)	0.056 (0.010)	0.042 (0.014)	0.021 (0.016)	−0.022 (0.018)	−0.084 (0.030)	−0.056 (0.044)	−0.11 (0.074)
Within R^2	0.78	0.77	0.75	0.74	0.72	0.70	0.70	0.66	0.61	0.65
Observations	263,527	276,795	256,020	256,637	266,652	259,741	260,926	259,466	257,506	259,821

[a]*Notes*: $\hat{\beta}^+$ and β^- as estimated from specification (2) for separate deciles of the wage, age, tenure, and team size distributions. We include observation i in the decile k in t if i falls into the kth decile of the pooled distribution. Team Definition 2 at horizon

Table IV: heterogeneity across wage, age, tenure, and team-size distributions.

5.3 Tables V–VII: switchers, managers, and occupation proximity

Read:

- Table V shows the high-paid coworker effect persists for establishment switchers, for switchers with nonemployment spells, and for switchers associated with mass-layoff events.
- Table VI shows manager and worker peer wages have very similar effects, suggesting the managerial label itself is not the key learning source.
- Table VII separates peers in the same occupation and other occupations. High-paid peers in the same occupation have the strongest effect.
- For same-occupation peers paid less than the worker, coefficients are near zero at short horizons.

Economic message: The effects are not purely establishment-specific and are strongest when coworker knowledge is more relevant to the worker's own tasks.

TABLE V

ESTABLISHMENT SWITCHERS[a]

Horizon in Years	Panel A: All Switchers 1	2	3	5	10
$\bar{w}^{+}$	0.092	0.17	0.20	0.25	0.35
	(0.018)	(0.022)	(0.024)	(0.027)	(0.032)
$\bar{w}^{-}$	−0.052	−0.031	−0.019	−0.0082	0.026
	(0.011)	(0.0099)	(0.012)	(0.014)	(0.022)
Within R^2	0.59	0.55	0.49	0.40	0.26
Observations	194,848	228,110	203,726	160,495	43,609

Horizon in Years	Panel B: Switchers With Nonemployment Spell 1	2	3	5	10
$\bar{w}^{+}$	0.094	0.19	0.20	0.24	0.31
	(0.020)	(0.019)	(0.021)	(0.023)	(0.032)
$\bar{w}^{-}$	−0.039	0.024	0.044	0.036	0.061
	(0.022)	(0.014)	(0.016)	(0.018)	(0.030)
Within R^2	0.37	0.46	0.39	0.31	0.19
Observations	21,084	72,223	68,781	57,331	16,224

Horizon in Years	Panel C: Switchers, Mass Layoff Event 1	2	3	5	10
$\bar{w}^{+}$	0.11	0.14	0.14	0.22	0.34
	(0.056)	(0.046)	(0.051)	(0.055)	(0.089)
$\bar{w}^{-}$	0.032	0.032	0.064	0.049	−0.016
	(0.069)	(0.042)	(0.048)	(0.054)	(0.11)
Within R^2	0.34	0.34	0.28	0.21	0.14
Observations	2264	5258	5453	4904	1545

[a]*Notes*: $\hat{\beta}^{+}$ and β^{-} as estimated from specification (2) on a sample of establishment switchers. Team Definition 2. Column titles indicate horizon h. Standard errors clustered at the establishment level. The regressions include current wage and fixed effects for age decile, tenure decile, gender, education, occupation, and year. Standard errors in parentheses.

Table V: establishment switchers.

TABLE VI

MANAGER AND WORKER PEERS[a]

	Horizon in Years 1	2	3	5	10
$\bar{w}_w$	0.062	0.090	0.12	0.16	0.21
	(0.0039)	(0.0058)	(0.0077)	(0.010)	(0.017)
$\bar{w}_m$	0.058	0.084	0.11	0.16	0.22
	(0.0036)	(0.0053)	(0.0076)	(0.010)	(0.019)
Within R^2	0.88	0.82	0.77	0.67	0.47
Observations	3,517,568	3,086,436	2,661,936	1,934,856	458,302

[a]*Notes*: Variables are weighted: To construct $\bar{w}_m$, we compute the mean wage of workers in managerial occupation(s) and then weight by the fraction of overall establishment employment of managers. Column titles indicate horizon h. Standard errors clustered at the establishment level. The regressions include current wage and fixed effects for age decile, tenure decile, gender, education

TABLE VII

PEERS IN THE SAME AND IN OTHER OCCUPATIONS[a]

	Horizon in Years 1	2	3	5	10
$\bar{w}^{+}_{\text{same occ}}$	0.12	0.16	0.21	0.29	0.39
	(0.013)	(0.019)	(0.026)	(0.037)	(0.048)
$\bar{w}^{-}_{\text{same occ}}$	−0.00039	0.0076	0.013	0.022	0.043
	(0.012)	(0.017)	(0.023)	(0.033)	(0.047)
$\bar{w}^{+}_{\text{other occ}}$	0.086	0.12	0.15	0.21	0.30
	(0.0068)	(0.011)	(0.015)	(0.022)	(0.037)
$\bar{w}^{-}_{\text{other occ}}$	0.038	0.054	0.076	0.11	0.15
	(0.0064)	(0.0089)	(0.012)	(0.018)	(0.027)
Within R^2	0.88	0.81	0.76	0.66	0.46
Observations	3,315,351	2,907,077	2,509,645	1,827,701	431,782

[a]*Notes*: $\hat{\beta}^{+}$ and β^{-} as estimated from specification (2) with separate coefficients for peers in the same occupation and in other occupations. Variables are weighted: To construct $\bar{w}^{+}_{\text{same occ}}$ ($\bar{w}^{+}_{\text{other occ}}$), we compute the mean wage of workers above in the wage distribution in the same (other) occupation(s) and then weight by the fraction of overall establishment employment the (other) occupation(s) accounts for. Column titles indicate horizon h. Standard errors clustered at the establishment level. The regressions include current wage and fixed effects for age decile, tenure decile, gender, education, occupation, and year. Standard errors in parentheses.

Tables VI–VII: occupation-based peer groups.

5.4 Figure 1: nonparametric coworker wage distribution

Read:

- The figure plots marginal effects of moving peers across relative wage bins.
- Low-paid peer bins have similar and small effects.
- Effects increase for peers above the worker in the wage distribution.
- The bottom panel shows the response accumulates over time.

Economic message: The coworker distribution matters in a monotone way: workers gain more from having more peers above them, but gains are less than proportional for very far-ahead coworkers.

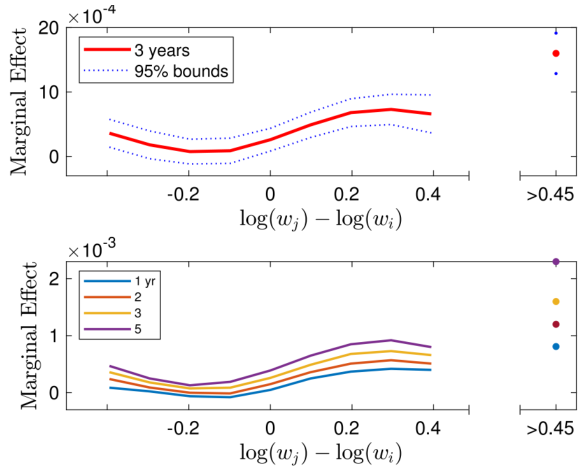

Figure 1: approximating the coworker wage distribution.

5.5 Tables VIII–IX: structural learning-function estimates

Read:

- Table VIII estimates the baseline learning function. For Team Definition 2, $\theta^+ = 0.0882$ and $\theta^- = 0.0370$.
- For Team Definition 1, $\theta^+ = 0.0673$ and $\theta^- = 0.0111$.
- The trend term θ^0 is small in both team definitions.
- Table IX shows that in the piecewise linear specification, $\theta^+ > \theta^{++}$: marginal learning from better coworkers is stronger when they are closer in knowledge.

Economic message: The structural estimates confirm the reduced-form result: learning from more knowledgeable coworkers is large, asymmetric, and strongest in occupation-specific teams.

TABLE VIII

PARAMETRIC ESTIMATION RESULTS FOR THE LEARNING FUNCTION (17)[a]

	Team Definition	
	1	2
θ^+	0.0673 (0.0004)	0.0882 (0.0006)
θ^-	0.0111 (0.0003)	0.0370 (0.0004)
θ^0	0.0039 (0.00004)	0.0060 (0.00003)
Observations	4,763,089	4,590,120

[a]GMM standard errors in parentheses.

Table VIII: baseline learning function.

TABLE IX

ESTIMATES FOR THE LEARNING FUNCTION (18)[a]

	Team Definition	
	1	2
θ^+	0.0844 (0.0011)	0.1091 (0.0009)
θ^{++}	0.0668 (0.0004)	0.0853 (0.0006)
θ^-	0.0102 (0.0003)	0.0346 (0.0004)
θ^0	0.0038 (0.00004)	0.0058 (0.00003)
Observations	4,763,089	4,590,120

[a]GMM standard errors in parentheses.

Table IX: piecewise linear learning.

5.6 Tables X–XII and Figure 2: quantitative implications

Read:

- Table X shows coworker learning is 8.27–8.68 percent of compensation under broad teams and 3.64–4.21 percent under narrow teams.
- Table XI shows random assignment would increase knowledge growth by about 64 percent under broad teams and 88–91 percent under narrow teams.
- Table XII shows inequality falls when learning is counted as compensation: about 33 percent under broad teams and 20 percent under narrow teams.
- Figure 2 studies team-size scaling. For narrow teams, the best fit occurs around $k = 0.8$, suggesting nontrivial returns to scale in learning.

Economic message: Learning is quantitatively important. Current team formation appears good for production but restrictive for knowledge diffusion, and wage inequality overstates compensation inequality.

TABLE X

COWORKER LEARNING AS A FRACTION OF COMPENSATION[a]

Learning Function	Team Definition 1	2
Basic Learning	8.27%	3.64%
Piecewise Linear Learning	8.68%	4.21%

[a]*Notes*: First row refers to results for learning function (17) and second row to results for learning function (18).

TABLE XI

INCREASE IN KNOWLEDGE GROWTH FROM RANDOM ASSIGNMENT[a]

Learning Function	Team Definition 1	2
Basic Learning	64.91%	91.14%
Piecewise Linear Learning	64.48%	88.07%

[a]*Notes*: First row refers to results for learning function (17) and second row to results for learning function (18).

TABLE XII

DECLINE IN INEQUALITY WHEN TAKING LEARNING INTO ACCOUNT[a]

Learning Function	Team Definition 1	2
Basic Learning	−33.47%	−19.56%
Piecewise Linear Learning	−33.53%	−19.70%

[a]*Notes*: We first compute the variance of log wages and then the variance of log overall compensation. We report the percentage decline when moving from the first to the second measure of inequality. First row refers to results for learning function (17); second row for learning function (18).

Tables X–XII: learning compensation, sorting, and inequality.

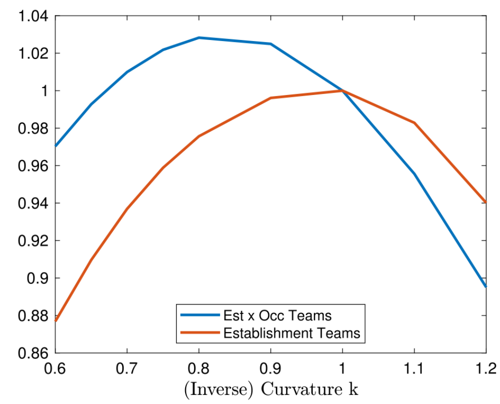

Figure 2: returns to team size.

6 Short summary

- The paper studies learning from coworkers using German administrative employer-employee data.

- Future wage growth is strongly associated with the wages of current coworkers.
- The relationship is much stronger for coworkers paid more than the worker.
- Effects are larger in establishment-by-occupation teams than in whole establishments.
- The effect persists for job switchers and mass-layoff movers, reducing concerns about back-loaded wages or firm-specific pay.
- Same-occupation high-paid peers have especially strong effects.
- A flexible bin specification shows wage growth rises with the share of coworkers above the worker in the wage distribution.
- The model interprets current wages as compensation for both current productivity and future learning opportunities.
- By choosing knowledge units as expected present value of earnings, the authors recover latent knowledge from wages and teams.
- Structural estimates show strong asymmetric learning from higher-knowledge coworkers.
- Coworker learning represents roughly 4–9 percent of total compensation.
- Random sorting would substantially raise knowledge growth, implying current team formation limits knowledge diffusion.
- Accounting for learning reduces measured inequality because low-knowledge workers receive more compensation as learning rather than current wages.

7 Conclusion

7.1 Core conclusion

Jarosch, Oberfield, and Rossi-Hansberg show that coworkers are an important source of learning and human-capital accumulation. Future wage growth is not only related to a worker's own wage and observables, but also to the wage distribution of coworkers, especially coworkers above the worker in the wage distribution.

7.2 Economic intuition

The intuition is that a job pays in two currencies: current wages and knowledge. A job with more knowledgeable coworkers may pay less today but more in the future through learning. In a competitive labor market, wages adjust to reflect this tradeoff, which allows the authors to infer hidden knowledge flows from wages and team composition.

7.3 Takeaway

The paper turns coworker interactions into a measurable form of compensation. It shows that learning at work is large, asymmetric, and shaped by team formation. This matters for wage growth, inequality, productivity, and the organization of production.